

LUKAS KLEIN

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WORK EXPERIENCE

08/2021 – Today
German Cancer Research Center (DKFZ)
Ph.D. Researcher in the Interactive Machine Learning Group supervised by Dr. Jäger and Prof. Maier-Hein.
- Machine Learning (ML) model training, MLOps, and evaluation within multi-GPU HPC environments.
- Leading ML work packages in interdisciplinary research projects (Biomedical, Material Science, Natural Language).
- Developing computer vision, sequence, tabular, and vision-language (generative) models in Python (PyTorch).
- Publications at: NeurIPS, ICLR, Advanced Materials (IF:30).

02/2021 – 08/2021
ETH Zürich
Research Intern and Master Thesis in the Institute for Machine Learning supervised by Prof. Buhmann.

01/2019 – 08/2019
Camelot ITLab
Associate IT Consultant for Machine Learning and cloud solutions in the pharma and chemical industry.

01/2018 – 01/2019
Camelot ITLab
Bachelor Thesis and Working Student for Data Science consulting.

06/2017 – 09/2017
Microsoft
Intern at the National Technology Office for AI in the public sector.

EDUCATION

2021 – Today
[Ph.D.] ETH Zürich
Doctor of Science in Computer Science at the Institute for Machine Learning supervised by Prof. Buhmann and Prof. Krause. External group member in the IVIA Lab of Prof. El-Assady. External collaboration with the MIT-IBM Watson AI Lab supervised by Dr. Strobelt.

2021 – Today
Helmholtz International Graduate School for Cancer Research
DKFZ International Ph.D. Program 3-year scholarship award with an acceptance rate of 2.7% in 2022.

2022
University of Oxford
OxML Summer School for Machine Learning in Healthcare. In person participation.

2019 – 2021
[M.Sc.] University of Geneva
M.Sc. in Statistics. Final grade: Very Good. Master Thesis and Exchange Semesters at ETH Zürich. Exchange grade: Very Good.

2015 – 2018
[B.Sc.] University of Mannheim
B.Sc. in Economics with minor subject Business Studies. Final grade: Good. Exchange at the Norwegian School of Economics. Exchange grade: Very Good.

SKILLS

Languages:

German	Native
English	Fluent
Chinese (Hanyu Pinyin)	Basic

Programming:

Python	Expert
R	Expert
Bash	Advanced
Julia	Basic
SQL	Advanced
CUDA	Basic
LaTeX	Expert
PyTorch	Expert
TensorFlow	Basic

Software & Platforms:

Linux (Ubuntu, CentOS, Debian)	Expert
Git (with GitHub & GitLab)	Advanced
Weights & Biases	Expert
Docker/Apptainer Container	Advanced
IBM Spectrum LSF	Expert
Nvidia Jetson IoT Platform	Advanced
CMake	Basic
NCI Genomic Data Commons & TCGA	Basic
Microsoft Power BI	Basic
Microsoft Visual Studio Code	Expert
Azure (Machine Learning Studio)	Advanced
Google Cloud (Cloud Build & Run, Vertex AI)	Advanced
SAP Analytics Cloud	Advanced

VOLUNTARY

2022 – Today **Advisory Committee of Oxford ML Summer School**
AI for Global Goals.

2022 – Today **Guest Lectures / Seminars**
TU Munich, DESY, HIDA Graduate School, HIDSS4Health Graduate School, IBM Zürich.

2022 – 2023 **Teaching Assistant**
Advanced Topics in Machine Learning (ETH Zürich).

2022 – 2023 **Conference Reviewer**
EuroVis 2022, ICML 2023.

2016 – 2017 **Speaker for Public Relations**
AStA (Student Council) at the University of Mannheim.

2015 – 2016 **Member of the Student Parliament and Deputy Member of the Information Committee**
University of Mannheim. Representative for Refugee Projects.

PUBLICATIONS

2024 **Explainable AI-based analysis of human pancreas sections detects traits of type 2 diabetes**
Authors: Lukas Klein, Sebastian Ziegler, Yanni Morgenroth, Eyke Schoeniger, Michele Solimena, Felicia Gerst, Robert Wagner, Fabian Isensee, Klaus Maier-Hein, Paul Jäger, **Venue:** *Under review at Nature Metabolism*.

2024 **Navigating the Maze of Explainable AI: A Systematic Approach to Evaluating Methods and Metrics**
Authors: Lukas Klein, Carsten Lüth, Udo Schlegel, Till Bungert, Mennatallah El-Assady, Paul Jäger,
Venue: *Under review at NeurIPS 2024*.

2024 **Enhancing predictive imaging biomarker discovery through treatment effect analysis**
Authors: Shuhan Xiao, **Lukas Klein**, Paul Jäger, Jens Petersen, Klaus Maier-Hein,
Venue: *Under review at Nature Scientific Reports*.

2024 **Interactive Semantic Interventions for VLMs: A Human-in-the-Loop Approach to Interpretability**
Authors: Lukas Klein, Kenza Amara, Carsten Lüth, Antonio Rodríguez, Hendrik Strobelt, Mennatallah El-Assady, Paul Jäger,
Venue: *Under review at NeurIPS 2024 Safe Generative AI Workshop*.

2024 **Explainable AI-based analysis of human pancreas sections detects traits of type 2 diabetes**
Authors: Lukas Klein, Sebastian Ziegler, Yanni Morgenroth, Eyke Schoeniger, Michele Solimena, Felicia Gerst, Robert Wagner, Fabian Isensee, Klaus Maier-Hein, Paul Jäger **Venue:** *Under review at NeurIPS 2024 IAI Workshop*.

2023 **Discovering Process Dynamic for Scalable Perovskite Solar Cell Manufacturing with Explainable AI**
Authors: Lukas Klein, Sebastian Ziegler, Felix Laufer, Charlotte Debus, Markus Götz, Klaus Maier-Hein, Ulrich W. Paetzold, Fabian Isensee, Paul Jäger, **Journal:** *Advanced Materials (IF: 30)* , Download here. This work has been covered i.a. by The Independent.

2023 **Understanding Solar Cell Manufacturing with Explainable AI**
Authors: Lukas Klein, Sebastian Ziegler, Felix Laufer, Charlotte Debus, Markus Götz, Klaus Maier-Hein, Ulrich W. Paetzold, Fabian Isensee, Paul Jäger, **Venue:** *NeurIPS 2023 Workshop on XAI in Action (Oral, Top 4 Paper)*.

2023 **Navigating the Pitfalls of Active Learning Evaluation: A Systematic Framework for Meaningful Assessment**
Authors: Carsten T. Lüth, Till J. Bungert, **Lukas Klein**, Paul Jäger,
Venue: *NeurIPS 2023*, Download here.

2023 **Semi-supervised automated Gleason Grading on WSI**
Authors: Maximilian Fischer, Lars Krämer, Sebastian Ziegler, **Lukas Klein**, Paul Jäger, Marco Nolden, Peter Neher, Fabian Isensee, Klaus Maier-Hein, **Venue:** *ECDP 2024 (Oral)*.

2023 **A Call to Reflect on Evaluation Practices for Failure Detection in Image Classification**
Authors: Paul Jäger, Carsten T. Lüth, **Lukas Klein**, Till J. Bungert,
Venue: *ICLR 2023 (Oral, Top 1%)*, Download here.

2022 **From Correlation to Causation: Formalizing Interpretable Machine Learning as a Statistical Process**
Authors: Lukas Klein, Mennatallah El-Assady, Paul Jäger,
Venue: *IJCAI 2022 Workshop on XAI*, Download here.

2022 **Improving Explainability of Disentangled Representations using Multipath-Attribution Mappings**
Authors: Lukas Klein, João B. S. Carvalho, Mennatallah El-Assady, Paolo Penna, Joachim M. Buhmann, Paul Jäger,
Venue: *MIDL 2022*, Download here.